# HOSPITAL BILLING SYSTEM

Python Programming Case Study

## 1. Problem Statement

Hospitals need a simple and accurate way to calculate a patient's bill from consultation, diagnostic and treatment charges. Manual calculations may cause errors when applying concessions and classifying payments.

## 2. Objective

To create a Python program that accepts patient details and charges, calculates the total, applies an eligible concession, finds the final bill and displays the payment category.

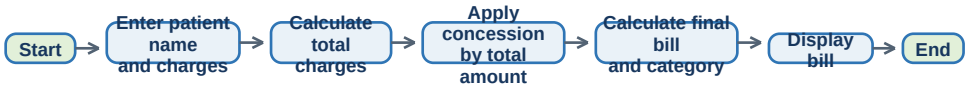
## 3. Assumptions

- Charges are numeric and non-negative.
- 10% concession for totals of INR 10,000 or more.
- 5% concession for totals of INR 5,000 or more but below INR 10,000.
- No concession below INR 5,000.
- Final-bill categories: High (INR 10,000 or more), Medium (INR 5,000 or more), otherwise Low.

## 4. Algorithm

1. Start and read the patient's name.
2. Read consultation, diagnostic and treatment charges.
3. Add the charges to calculate the total.
4. Apply 10%, 5% or 0% concession based on the total.
5. Subtract the concession to calculate the final bill.
6. Classify the final bill as High, Medium or Low Payment.
7. Display all bill details and stop.

## 5. Flowchart

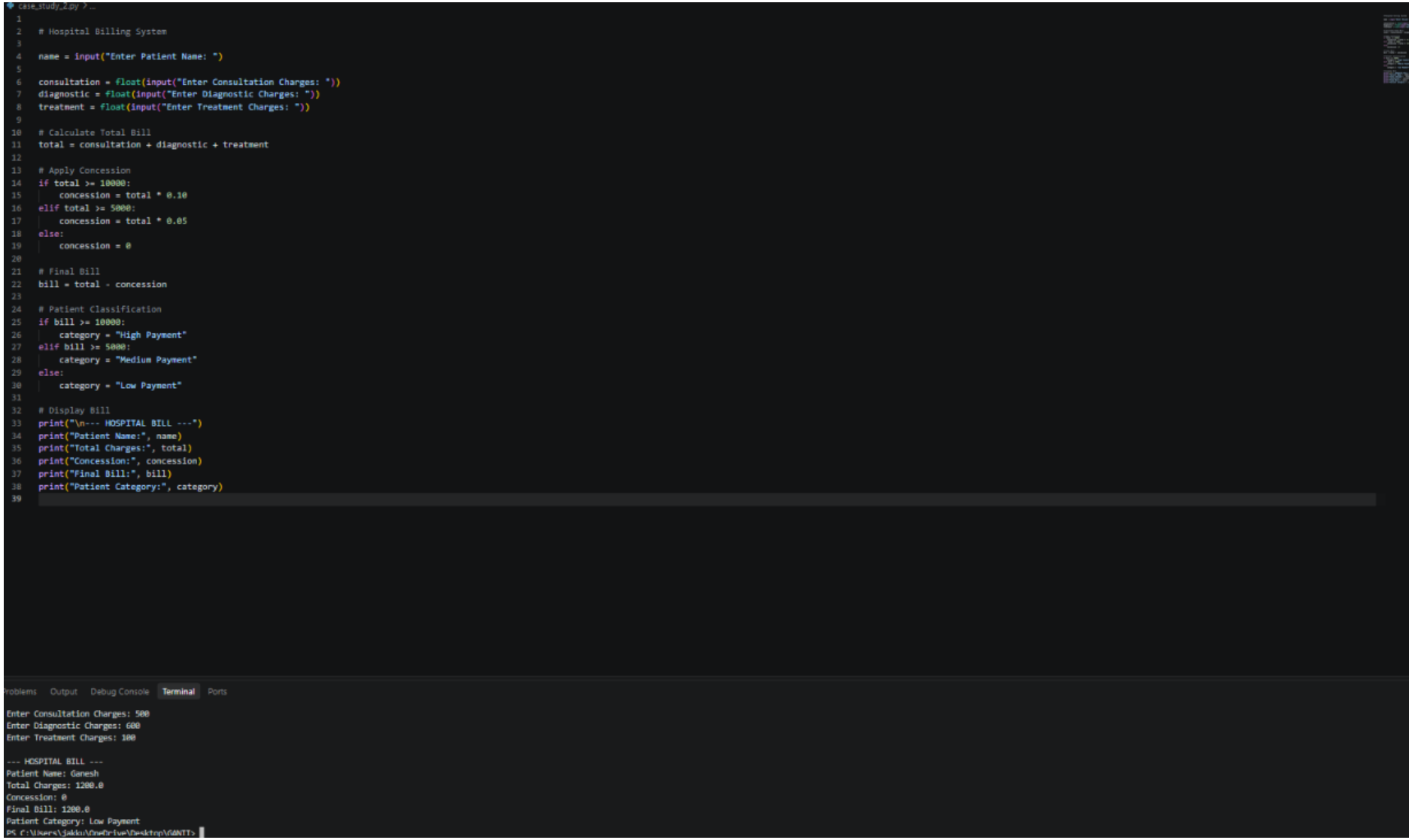



## 6. Pseudocode

**BEGIN**
INPUT name, consultation, diagnostic, treatment
total ← consultation + diagnostic + treatment
**IF** total ≥ 10000: concession ← total × 0.10
**ELSE IF** total ≥ 5000: concession ← total × 0.05
**ELSE:** concession ← 0
bill ← total – concession
**IF** bill ≥ 10000: category ← “High Payment”
**ELSE IF** bill ≥ 5000: category ← “Medium Payment”
**ELSE:** category ← “Low Payment”
DISPLAY name, total, concession, bill, category
**END**

# 7. Python Source Code and Output

Original screenshot supplied for this case study



```python
# Hospital Billing System

name = input("Enter Patient Name: ")

consultation = float(input("Enter Consultation Charges: "))
diagnostic = float(input("Enter Diagnostic Charges: "))
treatment = float(input("Enter Treatment Charges: "))

# Calculate Total Bill
total = consultation + diagnostic + treatment

# Apply Concession
if total >= 10000:
    concession = total * 0.10
elif total >= 5000:
    concession = total * 0.05
else:
    concession = 0

# Final Bill
bill = total - concession

# Patient Classification
if bill >= 10000:
    category = "High Payment"
elif bill >= 5000:
    category = "Medium Payment"
else:
    category = "Low Payment"

# Display Bill
print("\n--- HOSPITAL BILL ---")
print("Patient Name:", name)
print("Total Charges:", total)
print("Concession:", concession)
print("Final Bill:", bill)
print("Patient Category:", category)
```

```
Problems  Output  Debug Console  Terminal  Ports

Enter Consultation Charges: 500
Enter Diagnostic Charges: 600
Enter Treatment Charges: 100

--- HOSPITAL BILL ---
Patient Name: Ganesh
Total Charges: 1200.0
Concession: 0
Final Bill: 1200.0
Patient Category: Low Payment
```

## 8. Conclusion

The Hospital Billing System demonstrates Python input, arithmetic operations, conditional statements and output. It calculates total charges, applies the appropriate concession and classifies the final payment amount, making basic hospital billing clearer and more consistent.